PV101 CTS Cohort 6 Day 2 Student Packet

Electricity Basics, Multimeter Practice, Series and Parallel

Program	Cohort	Day	Date
PV101	CTS Cohort 6	Day 2	2026-07-07

Big Idea

- Solar PV turns light into DC electricity. We use measurements instead of guesses.

Electricity Words

- Voltage = pressure
- Current = flow
- Resistance = opposition
- Power = work being done

Units

- Voltage = volts = V
- Current = amps = A
- Resistance = ohms
- Power = watts = W

Main Formula

- Watts = Volts x Amps
- Amps = Watts / Volts
- Volts = Watts / Amps

AC vs DC

- DC = direct current: batteries, solar panels, many RV loads
- AC = alternating current: wall outlets and many home appliances

Series and Parallel

- Series = voltage adds
- Parallel = current adds
- Memory line: Series stacks voltage. Parallel stacks amperage.

PV System Path

- Solar panel -> charge controller -> battery -> loads
- Battery -> inverter -> AC loads when needed

Meter Practice Record

Object Tested	Meter Setting	Expected Reading	Actual Reading	What Did It Tell You?
AA battery	DC volts			
9V battery	DC volts			

12V source	DC volts			
Small PV panel in light	DC volts			
Small PV panel shaded/covered	DC volts			
Wire continuity	Continuity			

RV Solar Load Starter List

Load	AC/DC	Voltage	Watts or Amps	Notes
Water pump	DC	12V	Unknown	Verify hidden fuse and label
Interior lights	DC	12V	Unknown	Count fixtures
Fan	DC	12V	Unknown	Check label or measure
USB charging	DC	5V/12V	Unknown	Adapter losses
Outlet loads	AC	120V	Unknown	Inverter needed

Exit Ticket

One thing I understand better today:

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One thing I still want to practice:

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One safety rule I need to remember:

_____

One RV load we should verify before designing the system:

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